Transit Accessibility and Spatial Equity Analysis

Greater Portland, Maine | PostgreSQL/PostGIS, Python ETL, Spatial SQL, QGIS Automation
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TRANSIT STOPS	TRANSIT ROUTES	SIDEWALK SEGMENTS	NEIGHBORHOOD UNITS
982	**416**	**10,670**	**7**

Executive Summary

This project evaluates transit accessibility and spatial equity across a first-run Greater Portland analysis area. The pipeline downloads public GPCOG transit, route, sidewalk, and study-area layers, loads them into PostGIS, runs spatial SQL, and exports a QGIS map and report-ready findings.

The 800-meter transit stop buffer covers about 50,939 residents (69.12%) across the current neighborhood sample, leaving 22,761 residents outside that access band.

TOWNS	ANALYSIS UNITS	POPULATION	800 M COVERAGE	WEIGHTED SCORE
6	**7**	**73,700**	**69.12%**	**61.03**

Important interpretation note: neighborhoods, schools, and hospitals are currently synthetic sample records. They keep the end-to-end workflow operational while verified public Census and facility datasets are added. Treat the workflow, SQL, and cartographic automation as production structure, and treat current neighborhood-level scores as demonstration values until those layers are replaced.

KPI Questions Covered

- How many towns and analysis units are included in the first-run study area?
- What share of each town's population is estimated inside 400-meter and 800-meter stop-access bands?
- Which towns have the strongest population-weighted accessibility scores, and what component scores explain them?
- How far is each town from its nearest transit stop on average, and what is the longest nearest-stop distance inside that town?
- Which towns contain underserved analysis units under the current score and distance thresholds?
- Which dimension is the limiting factor for each analysis unit: transit, sidewalk, school, or hospital access?

All-Town KPI Dashboard

The tables below report every town represented in the current analysis units. Scores are population-weighted at the town level so larger units contribute proportionally to the town summary.

Town Coverage and Score Summary

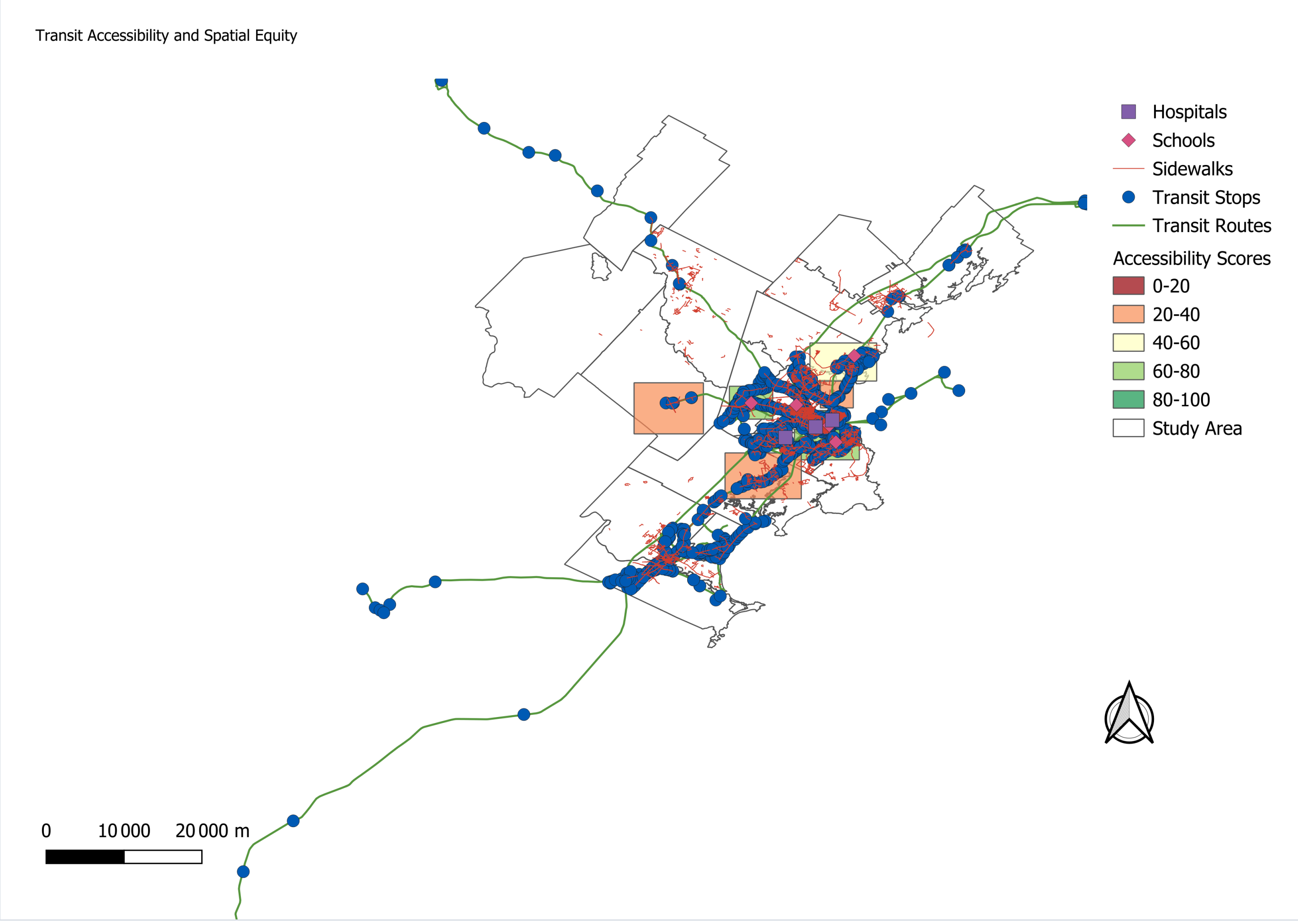
Rank	Town	Units	Population	Pop 400 m	Pct 400 m	Pop 800 m	Pct 800 m	Weighted Score	Avg Stop m	Max Stop m	Underserved Units
1	Portland	2	25,200	16,812	66.71%	22,555	89.50%	74.18	269.27	416.32	1
2	South Portland	1	13,500	7,670	56.81%	11,645	86.26%	72.27	77.41	77.41	0
3	Westbrook	1	11,800	5,794	49.10%	9,493	80.45%	64.44	39.46	39.46	0
4	Falmouth	1	6,400	1,566	24.47%	2,965	46.33%	52.13	169.19	169.19	0
5	Scarborough	1	9,000	1,453	16.14%	3,336	37.07%	36.46	716.90	716.90	1
6	Gorham	1	7,800	282	3.62%	946	12.13%	29.58	731.30	731.30	1

Town Component and Asset KPIs

Town	Transit	Sidewalk	School	Hospital	Sidewalk km	Sidewalk m/1k Pop	Schools	Hospitals	Units Within 800 m	Units Farther 800 m
Portland	85	55.85	34.62	37.01	187.47	7,439.14	1	2	2	0
South Portland	100	71.29	43.58	21.66	119.64	8,862.35	1	0	1	0
Westbrook	100	16.35	97.69	0	27.44	2,325.79	1	0	1	0
Falmouth	100	27.58	19.27	0	46.29	7,232.05	1	0	1	0
Scarborough	70	28.20	0	0	47.33	5,258.70	0	0	1	0
Gorham	70	5.25	0	0	8.82	1,130.60	0	0	1	0

Map Output

The map was generated from PostGIS layers using the QGIS Python API. It includes transit routes, stops, sidewalks, schools, hospitals, study-area boundaries, graduated accessibility-score polygons, legend, scale bar, and north arrow.



Interactive Dashboard Output

A standalone browser dashboard is generated at `reports/mobility_insecurity_dashboard.html`. It uses MapLibre GL JS, deck.gl, and Apache ECharts to support town filtering, map tooltips, score comparisons, coverage charts, component-score radar views, and a full all-town KPI table.

Spatial Analysis Results

Transit Coverage Buffers

Buffer	Population Inside	Population Outside	Percent Covered
400 m	33,577	40,123	45.56%
800 m	50,939	22,761	69.12%

All Analysis Unit KPI Detail

Town	Analysis Unit	Population	Pct 400 m	Pct 800 m	Nearest Stop m	Within 800 m	Score	Transit	Sidewalk	School	Hospital	Limiting Dimension	Category	Underserved
Falmouth	Falmouth Foreside	6,400	24.47%	46.32%	169.19	Yes	52.13	100	27.58	19.27	0	Hospital	Moderate	No
Gorham	Gorham Center	7,800	3.62%	12.13%	731.30	Yes	29.58	70	5.25	0	0	School	Low	Yes
Portland	East Deering	7,200	29.40%	64.18%	416.32	Yes	31.51	70	11.70	0	0	School	Low	Yes
Portland	Portland Peninsula	18,000	81.64%	99.63%	122.21	Yes	91.25	100	100	69.23	74.02	School	High	No
Scarborough	Scarborough North	9,000	16.14%	37.07%	716.90	Yes	36.46	70	28.20	0	0	School	Low	Yes
South Portland	South Portland Waterfront	13,500	56.81%	86.26%	77.41	Yes	72.27	100	71.29	43.58	21.66	Hospital	Moderate-high	No
Westbrook	Westbrook Core	11,800	49.11%	80.45%	39.46	Yes	64.44	100	16.35	97.69	0	Hospital	Moderate-high	No

Accessibility Scores

Neighborhood	Population	Accessibility	Transit	Sidewalk	School	Hospital	Nearest Stop m
Portland Peninsula	18,000	91.25	100	100	69.23	74.02	122.21
South Portland Waterfront	13,500	72.27	100	71.29	43.58	21.66	77.41
Westbrook Core	11,800	64.44	100	16.35	97.69	0	39.46
Falmouth Foreside	6,400	52.13	100	27.58	19.27	0	169.19
Scarborough North	9,000	36.46	70	28.20	0	0	716.90
East Deering	7,200	31.51	70	11.70	0	0	416.32
Gorham Center	7,800	29.58	70	5.25	0	0	731.30

Underserved Area Ranking

Rank	Neighborhood	Score	Nearest Stop m	Score Below 40	Farther Than 800 m
1	Gorham Center	29.58	731.30	Yes	No
2	East Deering	31.51	416.32	Yes	No
3	Scarborough North	36.46	716.90	Yes	No

Methods

The database uses geometry columns in EPSG:4326 and transforms geometries to EPSG:26919 for meter-based analysis. Transit stop buffers are created with **ST_Buffer** and dissolved with **ST_Union**. Population coverage is estimated by intersecting neighborhood polygons with 400-meter and 800-meter buffers. Nearest stop and facility distances are calculated from neighborhood centroids with projected **ST_Distance** measures and KNN-style ordering.

The accessibility score is a weighted index from 0 to 100: 40 percent transit access, 30 percent sidewalk access, 20 percent school access, and 10 percent hospital access. Underserved neighborhoods are flagged when the score is below 40, the nearest stop is more than 800 meters away, or the polygon is outside the 800-meter transit buffer.

Technical Summary

Layer	Records
Hospitals	3
Neighborhoods	7
Schools	5
Sidewalk segments	10,670
Study-area polygons	17
Transit routes	416
Transit stops	982

PostGIS version: POSTGIS="3.6.2 3.6.2" [EXTENSION] PGSQL="170" GEOS="3.14.1dev-CAPI-1.20.4" PROJ="8.2.1 NETWORK_ENABLED=OFF URL_ENDPOINT= USER_WRITABLE_DIRECTORY=C:\WINDOWS\ServiceProfiles\NetworkService\AppData\Local/proj" (compiled against PROJ 8.2.1) LIBXML="2.12.5" LIBJSON="0.12" LIBPROTOBUF="1.2.1" WAGYU="0.5.0 (Internal)" TOPOLOGY

Key SQL artifacts: `coverage_buffers`, `coverage_summary`, `nearest_transit_stop`, `neighborhood_sidewalk_access`, `accessibility_scores`, `underserved_areas`, `analysis_unit_accessibility_kpis`, `town_accessibility_kpis`, and `neighborhood_accessibility_map`.

Transit Accessibility and Spatial Equity Analysis for Greater Portland, Maine. Built with Python, PostgreSQL/PostGIS, spatial SQL, and QGIS automation.